The sun as colliding beam cosmic ray factory

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Talman full paper link

https://github.com/jtalman/ual2/blob/main/docs/

Sun-as-colliding-beam-cosmic-ray-factory.pdf

2 Abstract

A theory of cosmic ray production in the solar system

- Extra-galactic cosmic rays are needed only for initial start-up
- How can positrons and anti-protons be present even at very low and very high energies ?
- **Initially**, nuclear particles and electrons are captured by the sun's **magnetic** field, into counter-circulating low energy beams.
- **Later**, at higher energy, the **magnetic** field bending has become negligible compared to the **gravitational** bending;
- Both positive and negative beams survive the gradual transition from magnetic to gravitational bending.
- Aside: this requires a serious "**fine-tuning**" of the sun's magnetic bending field.
- Positrons and anti-protons are produced in collisions beween counter-circulating beam particles.

3 Abstract (cont)

- The only role for extra-galactic nuclear particles is to "start up" the solar process from scratch,
- Eventually the self-sustaining atmosphere of cosmic rays currently detectable within the solar system has been produced.
- Perhaps, but not necessarily, this start-up needs to be repeated every 11 years, coinciding with solar magnet and electric field reversals.
- Certainly the cosmic ray production must have been started up at least once in the past.
- The cosmic ray energy inherited from outside the solar system is negligiby small compared to the energy povided by the sun.

4 Abstract (cont)

- High quality cosmic ray data has neen collected over recent decades, at steadily increasing energies, especially by the International Space Station (ISS).
- Longitudinal electric fields, explained by the Parker theory of the solar wind, enable the sun to serve as a “booster” accelerator of cosmic rays, increasing cosmic ray energies by many orders of magnitude.
- High energy particle collision processes maintain the particle abundances at every energy.
- The following slides mainly show figures from the full paper.

5 Solar Accelerator

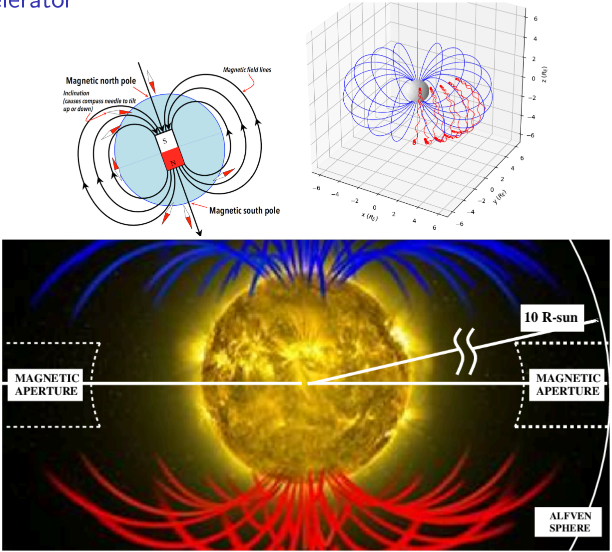

Figure 1: **Top:** Magnetic dipole field pattern views. **Bottom:** The sun's magnetic field lines, with the sun as a particle accelerator.

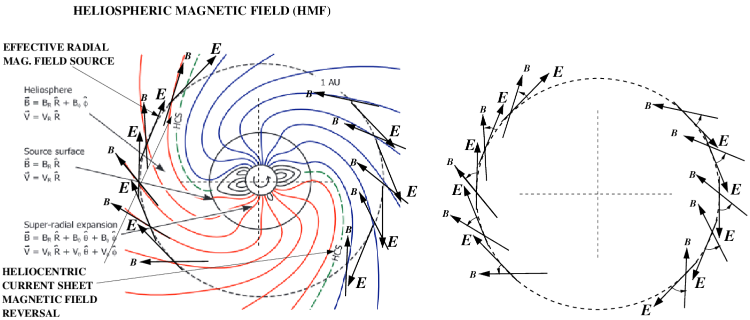

Figure 2: **Left:** Copied from Owens and Forsythe, solar wind field pattern, with **Right:** skeleton view of magnetic and electric field patterns.

- Magnetic field lines are plotted every 20 degrees.
- The electric field directions correspond to the Faraday's law electromotive force resulting from the 22 year period, time-varying axial solar "Kerst betatron" electric field pattern.

7 Alfven surface location determination

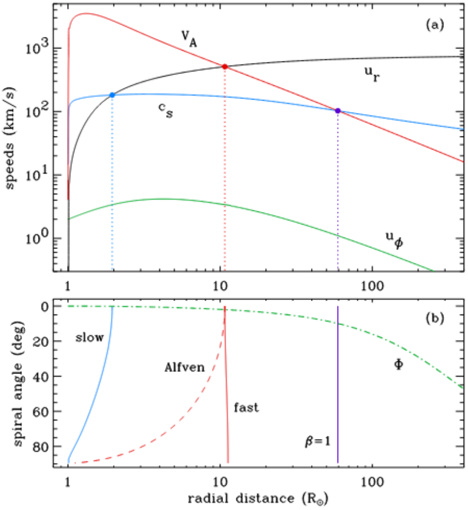

Figure 3: Figure copied from Crammer, calculations showing the (erratic) radius of the Alfven surface at aproximately 10 times the solar radius. β is not the ratio to the speed of light.

8 Experimental Apparatus (for thought experiment)

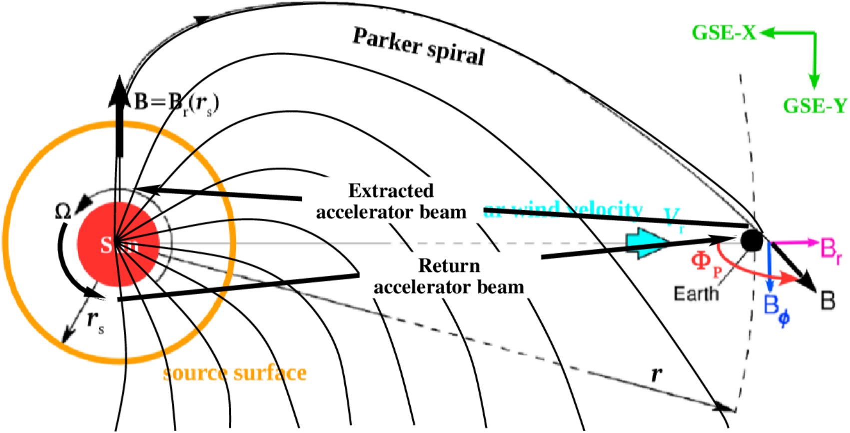

Figure 4: Experimental "Apparatus". The FNAL Tevatron beam is bent up by 5 degrees and pointed toward the sun. The protons are expected back in half an hour. But they don't show up? Precession of the elliptical orbit has not been taken into account.

9 Proton trip from and to a remote source

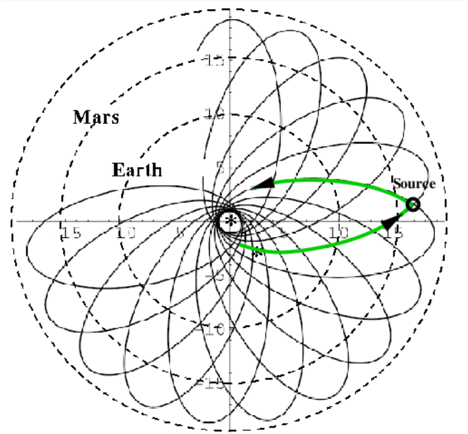

Figure 5: Stable bound orbit for a 6.6 GeV proton traveling **at velocity 0.99 c**, from and to a remote source. Figure copied and annotated from a paper by Munoz and Pavic, doi:10.1088/0143-0807/27/5/001.

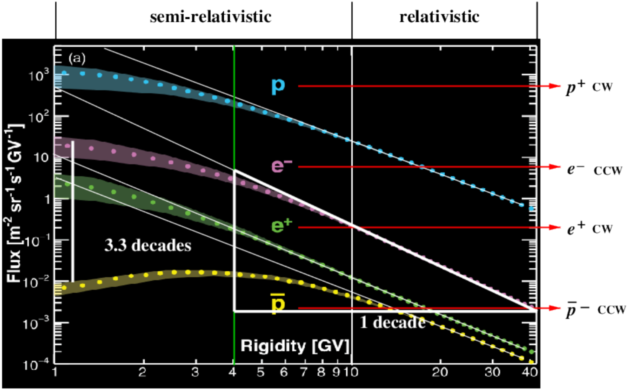

Figure 6: Figure copied from AMS paper reference[**?**], showing rigidity (i.e.. momentum over charge) dependence and relative abundances of protons, electrons, positrons, and anti-protons measured by the AMS specrometer on the ISS. $\bar{p}$'s and e^+ are produced by QED ineractions between counter-circulating beams. The protons are fully relativistic only above 10 GeV; electrons midway through the region labeled "semi-relativistic".

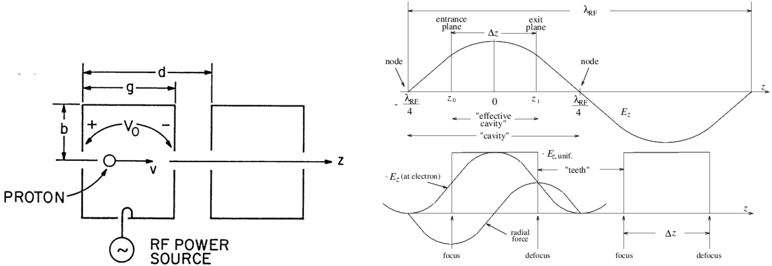

Figure 7: Copied from Loew and Talman, the **left** figure shows one of the many RF-cavity accelerating cells of an Alvarez linac needed to produce a "slow" (meaning slower than the speed of light) electromagnetic traveling wave. produce the desired wave speed, such that a moving proton is always "on crest". The **right** figure resembles Figure 12.1 in Kudsrud's "Plasma Physics for Astrophysics book".

12 Comparison of Kulsrud and Alvarez charge buckets

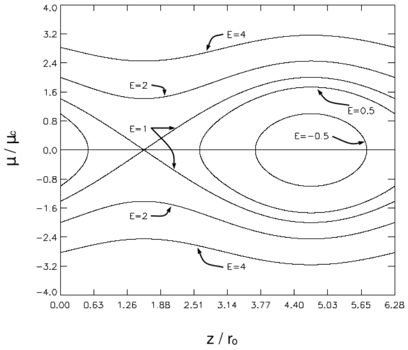

FIG. 1.—Contour plots of the particle energy. Note the presence of a separatrix at $E = 1$ and the trap region for $E < 1$.

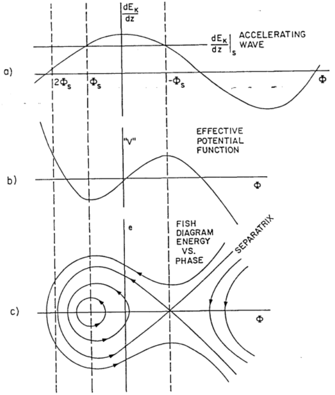

Fig. 6. Phase stability graphs.

Figure 8: **Left:** Copied from Felice and Kulsrud, longitudinal phase space in plasma atmosphere of a star resembles proton Alvarez linac figures on the right. **Right:** copied from Loew and Talman, longitudinal phase space in an "Alvarez" proton linac. These matching figures show that the physics of the acceleration of cosmic rays near a star can resembles the physics of acceleration of protons in a linear accelerator. In each case charges remain trapped in moving "stable buckets".

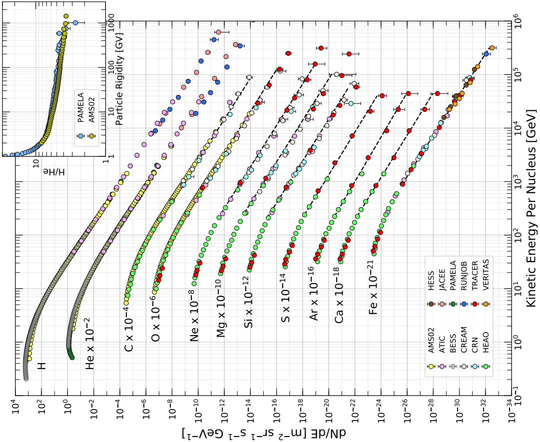

Figure 9: Figure copied from Particle Data Group, Chap-29, Cosmic-Rays. showing energy dependence and relative abundance ratios measured by AMS within the ISS. Some element abundances, such as the "light elements" (Li, Be, B) are not shown; presumably because these nuclei are so rare as cosmic rays.

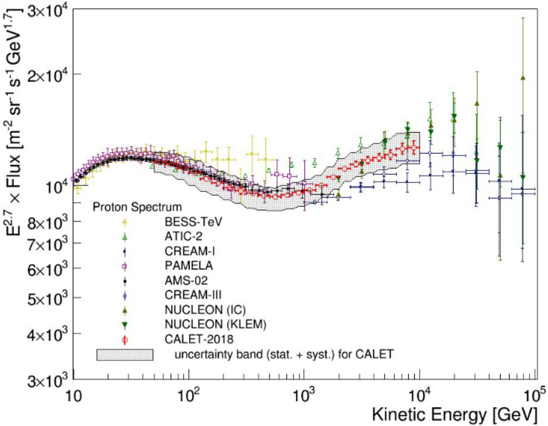

Figure 10: ISS measured flux of cosmic ray protons, plotted as function of proton energy: dN/dE flux dependence, with best fit exponential factor $\mathrm{KE}^{2.7}$ scaling;

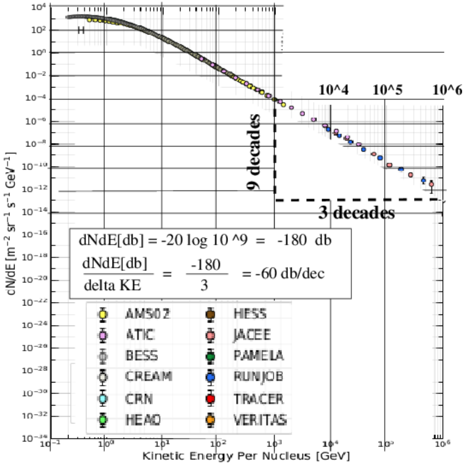

Figure 11: ISS measured flux of cosmic ray protons, plotted as function of proton energy: unscaled data, with best fit exponential decay. This slope of the eye-balled broken-line 3 decades per decade triangle matches the fully relativistic data, more or less consistent with the commonly accepted $\mathcal{E}^{2.7}$ energy scaling.

16 Variation of particle energy at fixed abundance ratio; ; i.e. “kink”

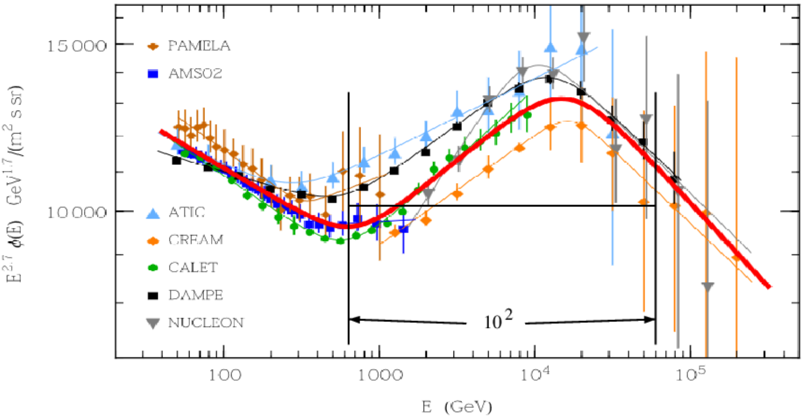

Figure 12: This figure has been copied from Lipari and Vernetto This data shows that all seven experiments on the ISS were subjected to the same impulse, or that all protons were subjected to the same impulse when near the sun. In the present paper this is interpreted as acceleration of protons from 600 GeV, to 60,000 GeV, while briefly circulating around the sun.

17 Variation of particle energy at fixed abundance ratio; i.e. "kink"

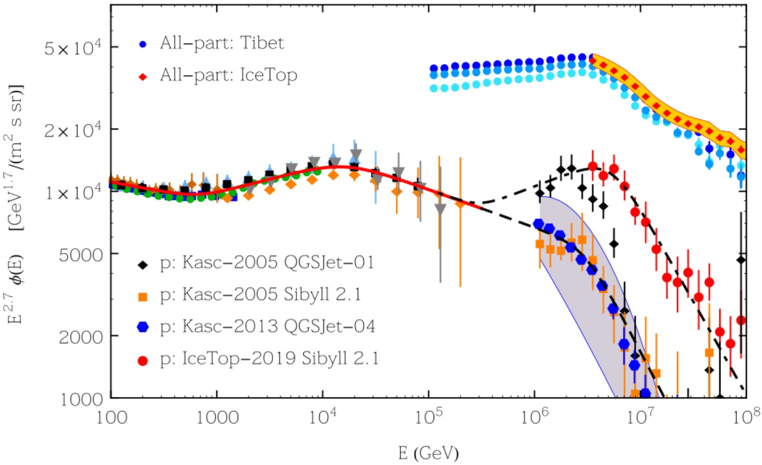

Figure 13: Also copied from Lipari and Vernetto the data in this plot increase the acceleration range, starting from 100 GeV and running almost to 10^7 GeV.

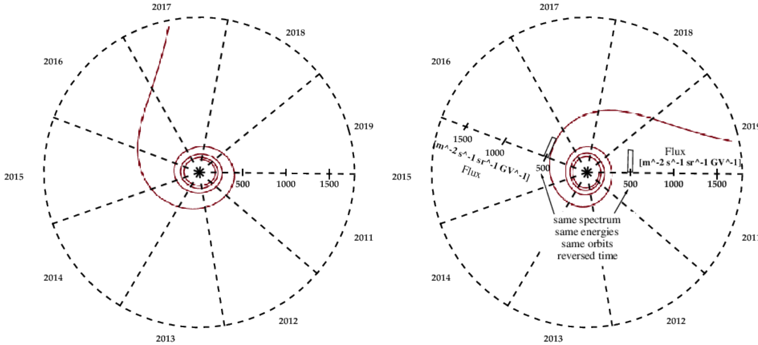

Figure 14: Calculated (pseudo)-orbits for a proton departing from AMS for the sun in mid 2017, or departing the sun towards the AMS. The spiral orbits are analytically calculated, lituus-modified spirals.

"Pseudo-orbits" from AMS aboard ISS

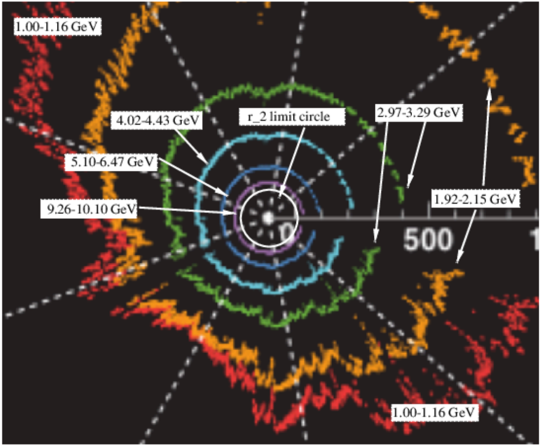

Figure 15: Cropped from M. Agular et al.; constructed from daily-sampled AMS proton fluxes for six rigidity bin counts from 1.0 to 10.10 GV, to form "pseudo-orbits" from daily data collected between 2011 and 2019. The data cover a significant fraction of solar cycle 24. On this radial scale the earth's orbit is at approximately 2000 units.